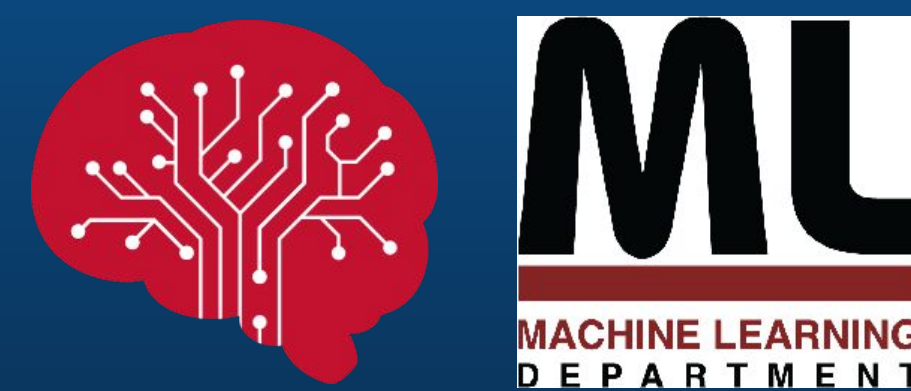


Eccentricity-Constrained CNN Training Reveals Adaptive Information Coding Around the Visual Field

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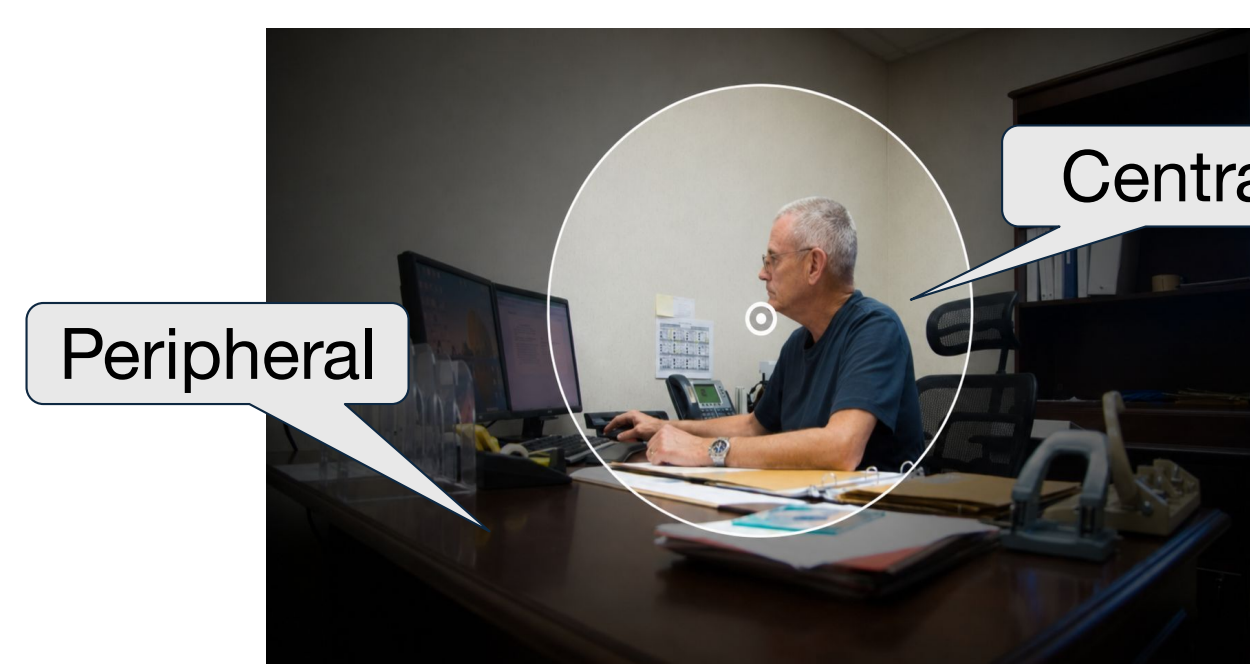
Read the paper!



Motivation

Visual-field eccentricity organizes visual cortex: face- and word-selective regions = center-biased, scene-selective regions = periphery-biased^{1,2}

Central vs. peripheral vision: central vision supports fine-detail recognition; peripheral vision supports global scene information^{3,4}



Question: Can these biases emerge in self-supervised CNNs trained on gaze-aligned, egocentric video?

Methods

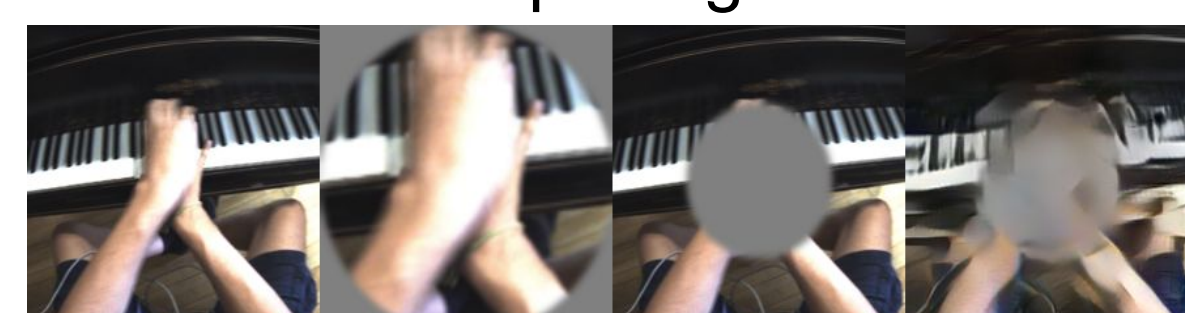
Visual Experience Dataset (VEDB)⁵ is a large egocentric video dataset with eye gaze. Matched VEDB frames were transformed into four gaze-contingent training conditions, used to pretrain otherwise identical SimCLR models⁶, and evaluated on visual recognition tasks as well as neural encoding with Natural Scenes Dataset (NSD)⁷.

VEDB Augments

Baseline Fovea-Gaze Periph Periph-NF



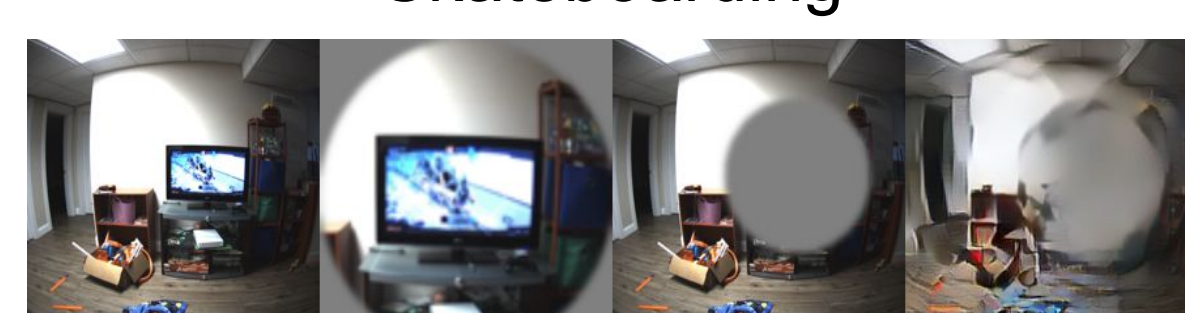
Exploring



Playing Piano



Skateboarding



Playing video games

Places365



VGGFace-2



Four matched training sets were created:

Baseline
Fovea-Gaze (Central)
Periph
Periph-NF (NeuroFovea)⁸

Frame identity, sample count, model architecture, and training budget were held constant.

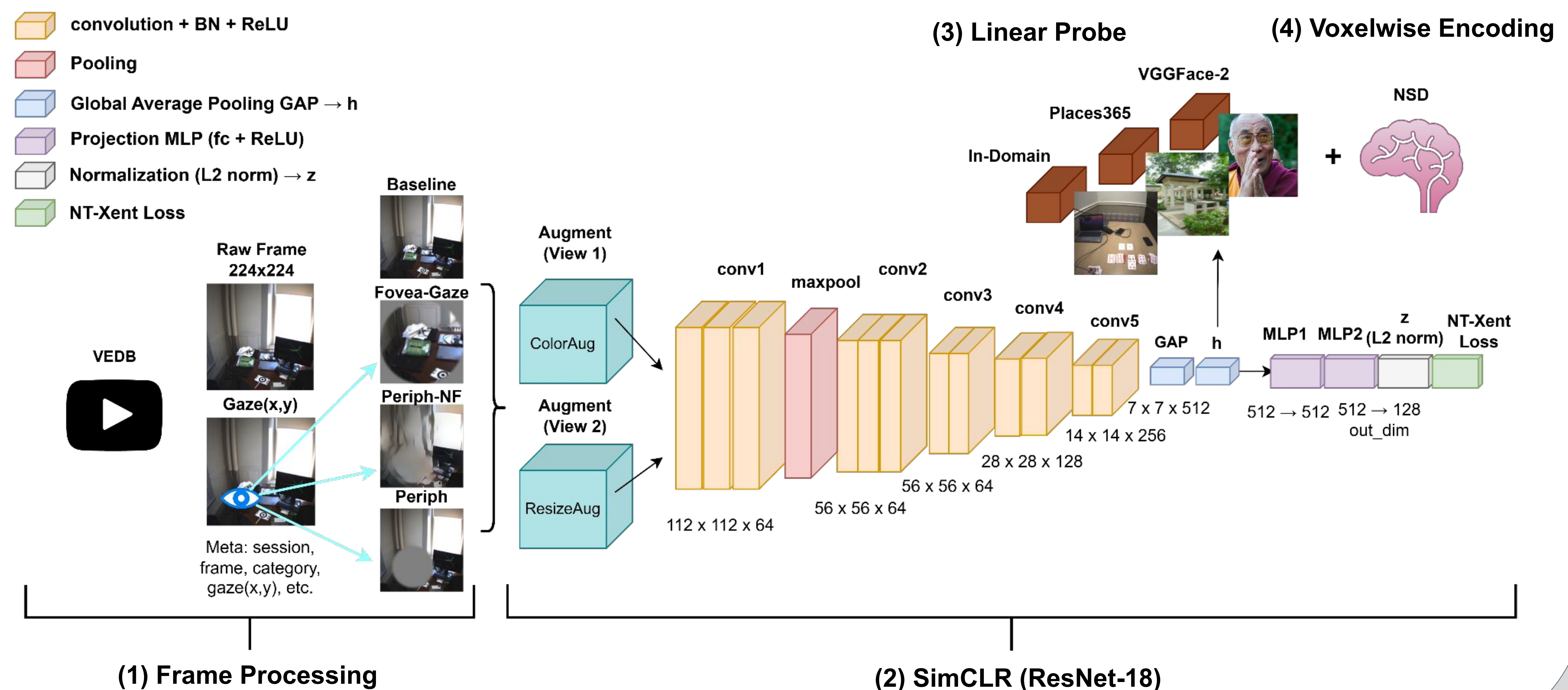
After SimCLR pre-training, frozen representations were evaluated on VEDB (In-Domain), Places365, VGGFace2, and NSD fMRI encoding.

Reference models included encoders pretrained on STL-10, ImageNet-100, and ImageNet-1K.

Analysis Details

- Four condition-specific SimCLR ResNet-18 models; 120 pretraining epochs
- Frozen-backbone linear probes on VEDB, Places365, and VGGFace2
- VEDB Core-17: labels with ≥ 500 frames across ≥ 7 sessions
- Session-disjoint train/validation/test splits to prevent temporal leakage
- In-domain performance evaluated primarily using class-balanced Macro-F1
- Category contrasts reported as $\Delta F1$ with 95% bootstrap CIs
- NSD voxelwise encoding evaluated using held-out R^2 and unique variance

Model Architecture

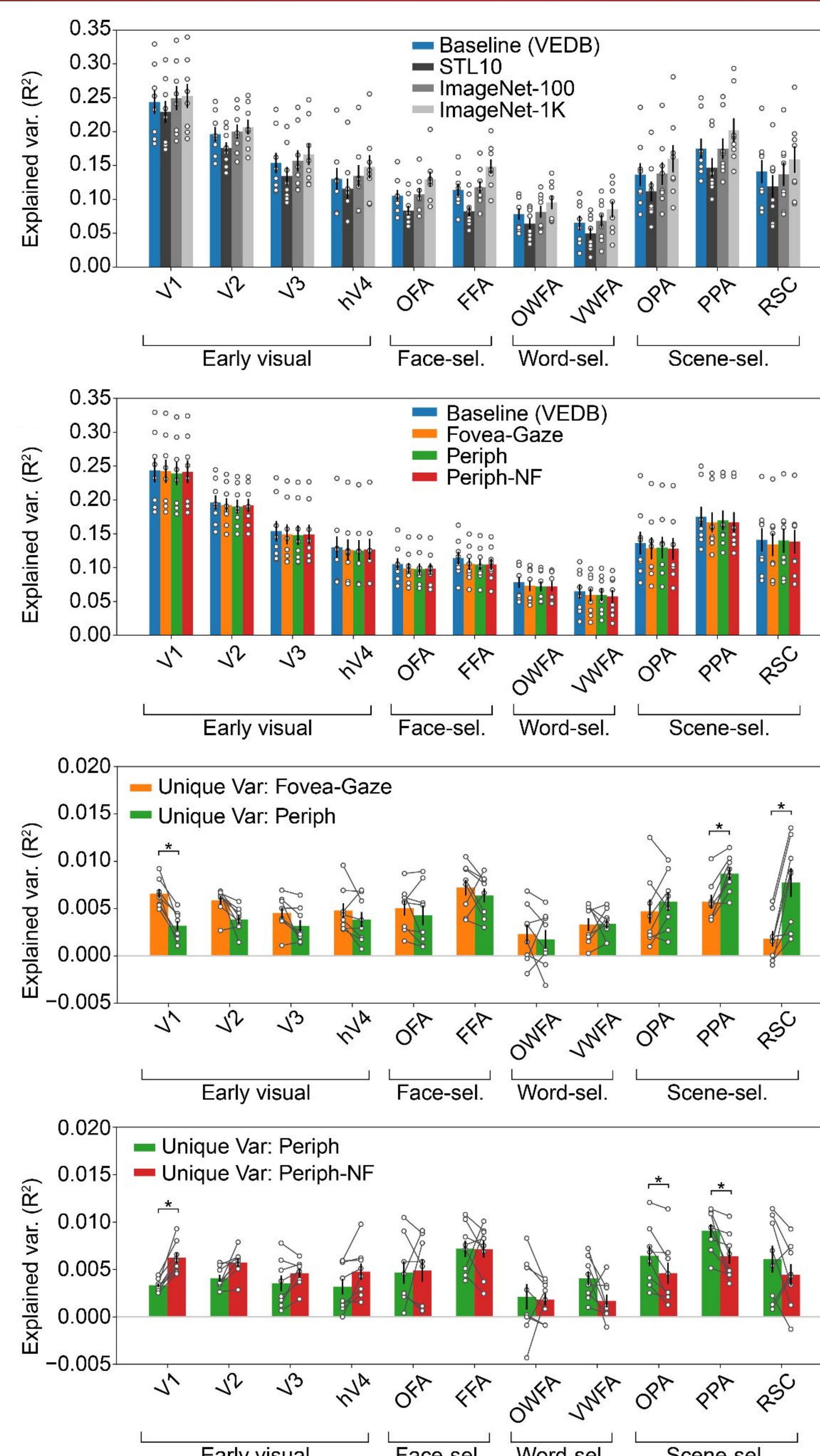


Downstream Evaluation

Classification Transfer: Fovea-Gaze performed best in-domain, whereas Baseline transferred best out-of-domain.

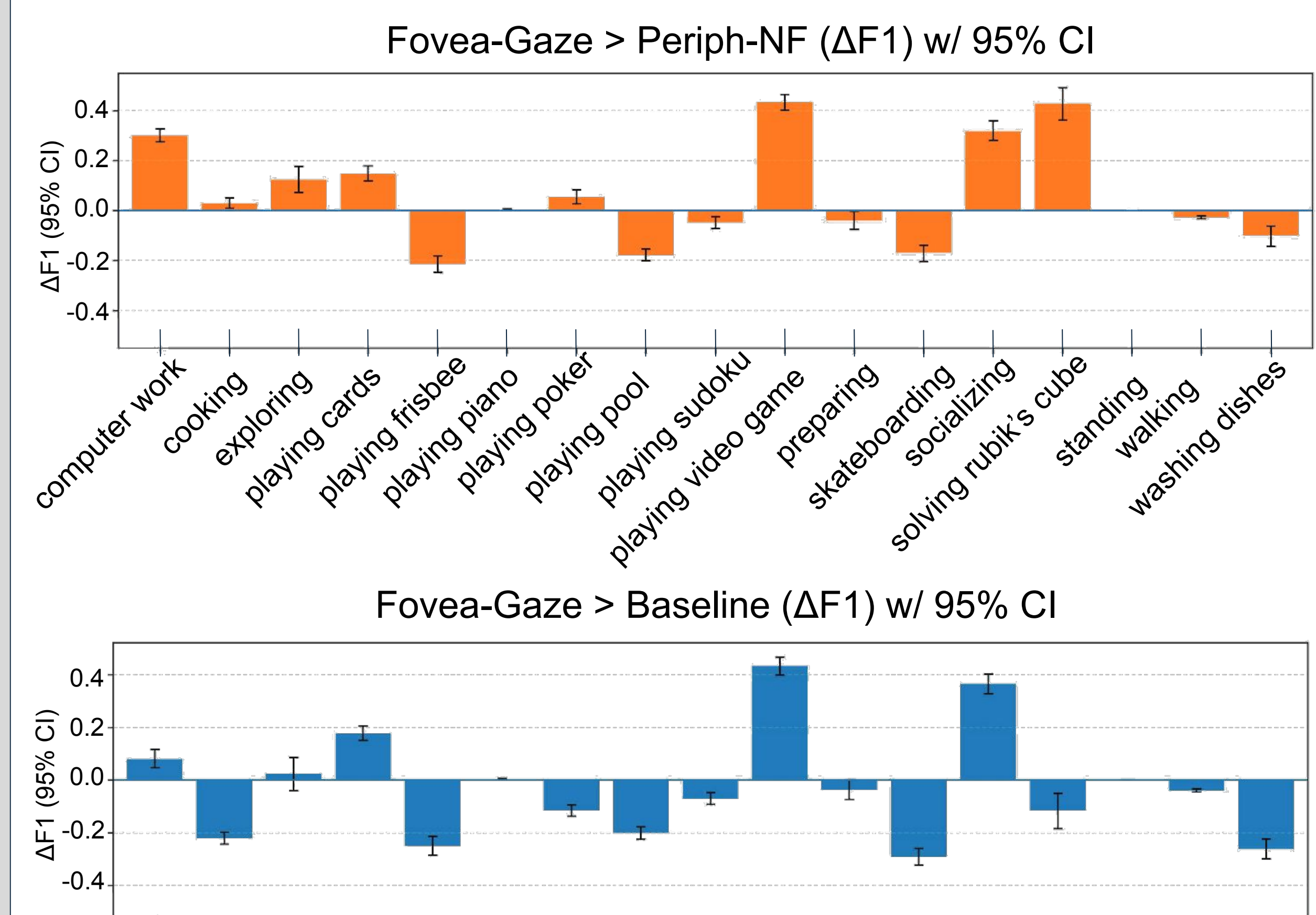
Pretraining	SimCLR Top-1 (%)	Downstream classification Macro-F1 (%)		
		In-Domain	VGGFace2	Places365
VEDB				
Baseline	87.60	42.17	3.26	23.16
Fovea-Gaze	90.43	43.64	2.70	19.14
Periph-NF	90.04	30.93	1.90	17.86
Periph	89.26	36.56	1.35	17.65
References				
STL-10	—	25.41	7.43	24.82
ImageNet-100	—	41.23	8.71	23.32
ImageNet-1K	—	43.33	8.77	28.36

Complementary Neural Alignment Across Visual Cortex in NSD



Systematic Differences Across Categories

Category-Specific Biases: Fovea-Gaze and Periph training favored different visual categories.



* $\Delta F1$ = Macro-F1 difference between conditions. Positive values favor the first named condition; A > B. Error bars show 95% bootstrap CIs.

Key Takeaways

- Eccentricity altered category-specific performance patterns**, indicating that central and peripheral inputs provide different task-relevant information.
- VEDB pretraining transferred more effectively to scenes than faces**, with foveal models outperforming peripheral models.
- Model differences aligned with cortical organization:** Periph favored scene-selective cortex, whereas Fovea-Gaze favored V1.
- VEDB pre-trained models were comparable to mid-sized non-egocentric models** in neural predictivity.

References & Acknowledgements

References:
¹Hasson et al. (2002); *Neuron*
²Levy et al. (2001); *Nature Neurosci.*
³Henderson & Hollingworth (1999); *Psychol. Sci.*
⁴Larson & Loschky (2009); *JOV*
⁵Greene et al. (2024); *JOV*
⁶Chen et al. (2020); *ICML*
⁷Allen et al. (2022); *Nature Neurosci.*
⁸Deza et al. (2019); *ICLR*

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